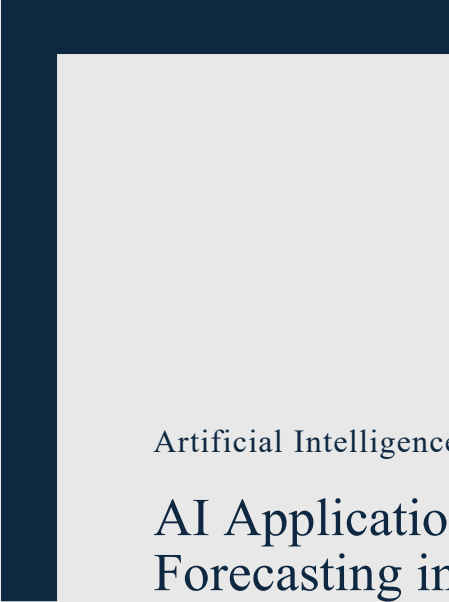




Artificial Intelligence for Sustainability (A25)

AI Applications for Short-Term Electricity Load Forecasting in Sustainable Energy Systems

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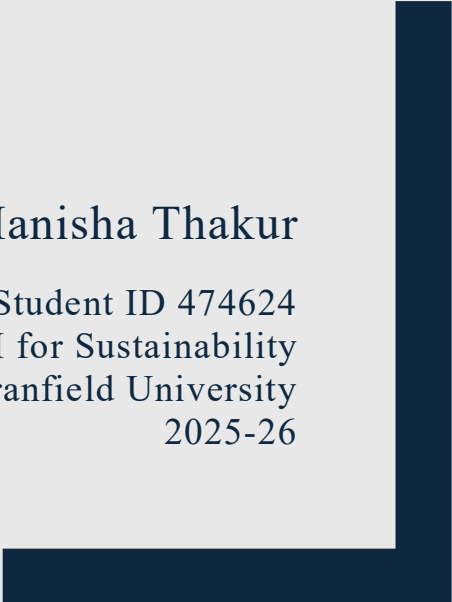


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Abstract

Artificial intelligence (AI) is the primary enabler of sustainable energy systems as increasing penetration of renewable energy sources introduces variability and uncertainty into electricity networks. Accurate forecasting plays the main role in maintaining system reliability, enabling flexibility, and reducing carbon emissions. This report reviews the state-of-the-art AI-based forecasting applications within the sustainability sector, with central focus on short-term electricity load forecasting. Literature about ensemble-based forecasting framework for buildings with a high proportion of renewable energy sources is analysed. Recent advances in deep learning, intelligent load forecasting technologies, and data-centric optimisation strategies are reviewed to provide broader methodological context. A practical ensemble forecasting implementation is then presented as an applied case study to demonstrate real-world deployment of AI-based forecasting. The report lists aims and objectives, methodologies, findings, and benefits to academia and industry, and concludes with recommendations for future research and practical implementation. The analysis demonstrates that ensemble AI forecasting approaches are best suited to renewable-rich environments and represent a robust and deployable solution for sustainable power system operation.

1. Introduction

The global transition toward sustainable and low-carbon energy systems has fundamentally transformed the operation of electricity networks. Large-scale deployment of renewable energy sources, particularly wind and solar photovoltaics, is essential for reducing greenhouse gas emissions and curbing climate change. Yet, the inherent intermittency and uncertainty of renewable generation introduce significant operational challenges for power systems. At the same time, electricity demand patterns are becoming increasingly complex due to the electrification of transport, heating, and industrial processes.

Accurate forecasting is essential for addressing these challenges. Short-term electricity load forecasting (STLF), typically covering time horizons from minutes to several days ahead, directly influences operational decisions such as unit commitment, reserve scheduling, and balancing market participation. Forecasting errors can lead to inefficient scheduling, increased reliance on carbon-intensive backup generation, and higher operational costs. As a result, forecasting accuracy is closely linked to both economic efficiency and sustainability outcomes.

Traditional forecasting approaches based on linear regression and classical time-series models often struggle to capture the nonlinear and non-stationary behaviour of modern electricity demand. Artificial intelligence (AI) and machine learning (ML) techniques have therefore gained increasing attention as powerful alternatives. AI-based models can learn complex temporal patterns from large datasets and adapt to fluctuating system conditions, making them useful for renewable-rich power systems.

This report focuses on forecasting as a key AI application in the sustainability sector. The objectives are to:

- (i) review the state of the art in AI-based short-term electricity load forecasting;
- (ii) analyse an ensemble-based forecasting framework as a representative application;
- (iii) demonstrate a practical implementation of AI-based forecasting through a case study;
- and
- (iv) evaluate benefits and provide recommendations for academia and industry.

2. State-of-the-Art Review of AI Forecasting in Sustainable Energy Systems

2.1 Forecasting as an AI Application in Sustainability

Forecasting is one of the most widely adopted AI applications in sustainable energy systems. It supports essential functions such as demand–supply balancing, renewable integration, and energy flexibility management. In renewable-rich environments, accurate forecasts enable system operators to reduce reserve margins, minimise renewable curtailment, and improve overall system efficiency (Liu et al., 2026).

Short-term load forecasting is mainly important because it directly affects real-time operational decisions. In building-level and local energy systems, accurate demand forecasts support optimal scheduling of storage systems and demand response programmes, thereby improving sustainability performance.

2.2 Evolution from Single Models to Ensemble Learning

Early AI-based load forecasting studies focused on individual models such as artificial neural networks, support vector machines, and decision trees. While these models improved forecasting accuracy compared with statistical approaches, their performance was often sensitive to data characteristics and hyperparameter choices. No single model consistently performed best across all operating conditions.

Ensemble learning has emerged as an effective solution to this limitation by combining multiple models to improve robustness and generalisation. By aggregating diverse learners, ensemble approaches reduce variance and mitigate individual model weaknesses. Liu et al. (2026) identify ensemble and hybrid frameworks as a key direction for intelligent load forecasting in new power systems, particularly under diverse and dynamic operating scenarios.

2.3 Deep Learning and Data-Centric Forecasting Advances

Recent research has mainly focused on deep learning architectures for short-term load forecasting. Dong et al. (2025) provide a comprehensive survey of deep learning approaches,

including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and long short-term memory (LSTM) networks. These models demonstrate strong performance by capturing nonlinear relationships and long-term temporal dependencies in electricity demand data.

In the meantime, data-centric approaches have gained attention. Cai et al. (2026) propose a synergistic cyclic optimisation strategy that iteratively integrates data screening and forecasting. Their results show that improving data quality can significantly enhance forecasting accuracy for electricity load as well as renewable generation. Although computationally intensive, such approaches highlight the importance of robust data preprocessing in AI-based forecasting systems. Table 1 compares forecasting performance of different models.

Table 1

Comparison of forecasting performance across individual models and the ensemble approach

Model category	Typical methods	Strengths	Limitations	Suitability for green power systems
Traditional ML (Good)	Linear regression, SVM	Simple, interpretable, low computational cost	Poor handling of nonlinearity and variability	Low-moderate renewable penetration
Deep learning (Better)	LSTM, GRU, CNN-LSTM	Captures complex temporal patterns, high accuracy	Data-hungry, less interpretable	Short-term solar, wind, and load forecasting
Hybrid / ensemble AI (Best)	Gradient boosting + RF, CNN-LSTM ensembles, cyclic optimisation	Robust, accurate, adaptable to uncertainty	Higher complexity	Renewable-rich and sustainable power systems

3. Core Academic Case Study: Ensemble-Based Load Forecasting for Renewable-Rich Buildings

3.1 Aims and Objectives

The core literature analysed in this report is the ensemble-based short-term load forecasting framework proposed by Pramanik et al. The primary aim of the study is to develop a robust forecasting approach capable of accurately predicting electricity demand in buildings with a high proportion of renewable energy sources. The specific objectives are to:

1. Improve forecasting accuracy compared with single-model approaches.
2. Enhance robustness under high variability caused by renewable integration.
3. Demonstrate the suitability of ensemble learning for sustainable building energy systems.

3.2 Methodological Framework

Pramanik et al. propose an ensemble-based forecasting framework that integrates multiple machine learning models to predict short-term electricity demand. The methodology follows a structured pipeline comprising data preprocessing, feature engineering, model training, and ensemble aggregation (Pramanik et al., 2024).

Historical load data is cleaned and normalised to ensure consistency and stability. Feature engineering includes temporal indicators such as hour of day, day of week, and seasonal variables, combined with lagged demand values and rolling statistics. These features enable the models to capture both short-term fluctuations and longer-term consumption patterns.

The ensemble combines predictions from multiple base learners using weighted averaging. Model weights are selected based on validation performance, allowing stronger models to contribute more heavily while retaining the stabilising influence of simpler learners. This design improves robustness and generalisation compared with individual models.

3.3 Key Findings

The results reported by Pramanik et al. demonstrate that the ensemble-based approach consistently outperforms individual forecasting models across multiple evaluation scenarios. Improvements in forecasting accuracy are particularly evident during periods of high demand variability, which are common in buildings with significant on-site renewable generation.

These findings confirm that ensemble learning is well suited to renewable-rich environments and align with broader conclusions in the intelligent load forecasting literature (Liu et al., 2026; Dong et al., 2025).

4. Applied Case Study: Practical Implementation of an Ensemble AI Forecasting Model

Flowchart of the Ensemble-Based Short-Term Electricity Load Forecasting Methodology

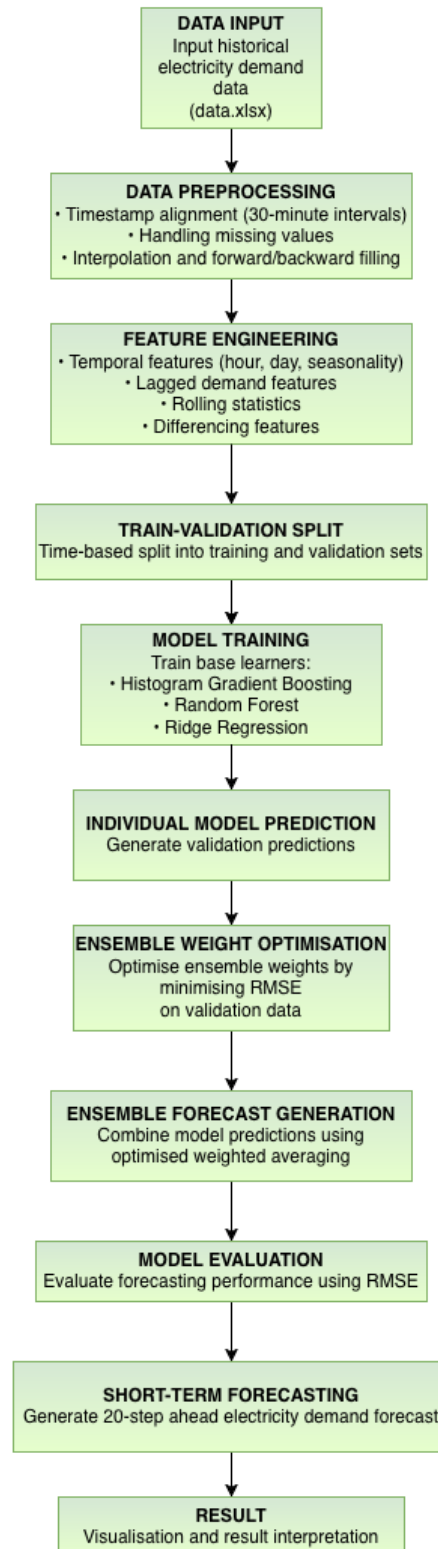


Figure 1

Flowchart of the Ensemble-Based Short-Term Electricity Load Forecasting Methodology

4.1 Application Context and Objectives

To complement the academic literature, a practical AI-based forecasting application was developed and documented in *Appendix B*. This applied case study demonstrates how ensemble forecasting concepts can be implemented in practice to support sustainable power system operation.

The objectives of the applied case study are to:

- (i) design a realistic forecasting pipeline that avoids data leakage;
- (ii) implement an ensemble machine learning model for short-term load forecasting; and
- (iii) evaluate forecasting performance using appropriate error metrics.

4.2 Data Processing and Feature Engineering

The applied model uses historical half-hourly electricity demand data spanning multiple years. Data preprocessing includes timestamp alignment, handling missing values, and interpolation of short gaps. A time-based split is used to separate training and validation datasets, reflecting real-world forecasting conditions and preventing information leakage.

Feature engineering includes temporal features (hour, day of week, month), cyclical encodings, lagged demand values, rolling statistics, and differencing features. All features are constructed using only past information to ensure operational feasibility.

4.3 Ensemble Model Design

The ensemble forecasting framework integrates three base learners: a histogram-based gradient boosting regressor, a random forest regressor, and a ridge regression model. Each model captures different characteristics of electricity demand, and predictions are combined using weighted averaging.

This ensemble design reflects principles discussed in the literature, where combining diverse models improves robustness and accuracy in fluctuating renewable-rich environments (Pramanik et al., Year; Liu et al., 2026).

4.4 Results and Performance Evaluation

Model performance was evaluated using root mean square error (RMSE) on a time-based validation set. Three individual models such as histogram-based gradient boosting (HGB), random forest (RF), and ridge regression were compared with an ensemble model combining the tree-based learners.

The ensemble model achieved the best performance with an RMSE of 11.03, outperforming all individual models. Among standalone models, HGB produced the lowest error (RMSE = 11.20), followed closely by RF (RMSE = 11.33). Ridge regression performed substantially worse (RMSE = 14.19), indicating its limited ability to capture nonlinear demand patterns.

Tree-based models demonstrated superior performance due to their ability to model complex temporal relationships and nonlinear interactions within the data. The ensemble approach further improved robustness by reducing variance and leveraging complementary model strengths.

As shown in Figure 2, the ensemble model closely tracked observed electricity demand across the validation period, accurately capturing daily load cycles and short-term fluctuations.

Figure 3 further demonstrates the practical application of the trained model by presenting a 20-step ahead electricity demand forecast, illustrating its suitability for short-term operational decision-making.

Residual diagnostics indicate that forecasting errors were generally centred around zero, with larger deviations occurring primarily during periods of rapid demand change (see Appendix A, Figures A1 and A2). Overall, the results confirm that ensemble learning provides an accurate and reliable solution for short-term electricity load forecasting in sustainable energy systems.

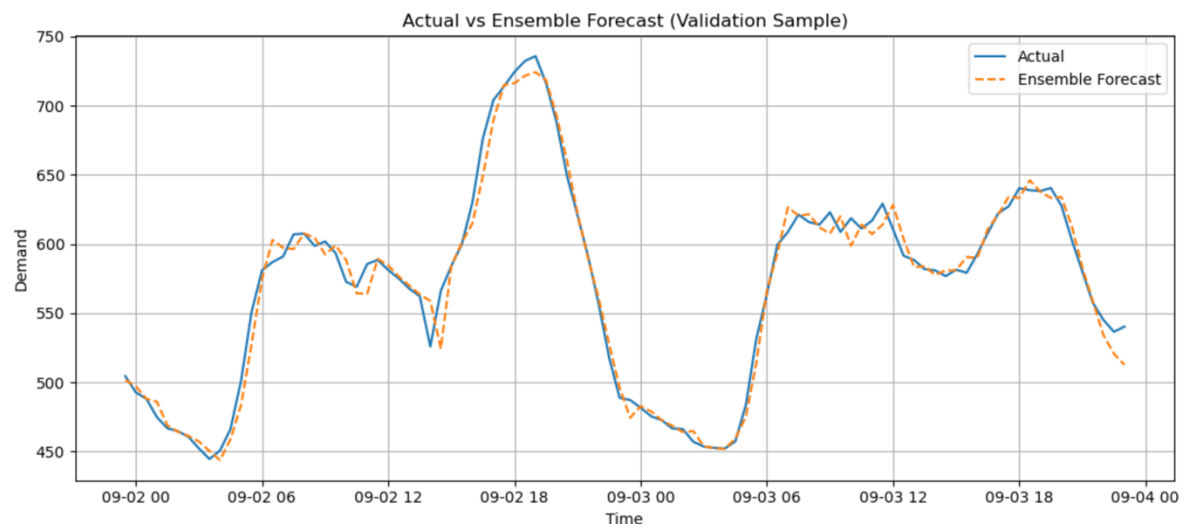


Figure 2

Comparison of actual and ensemble-forecast electricity demand over a sample of the validation period.

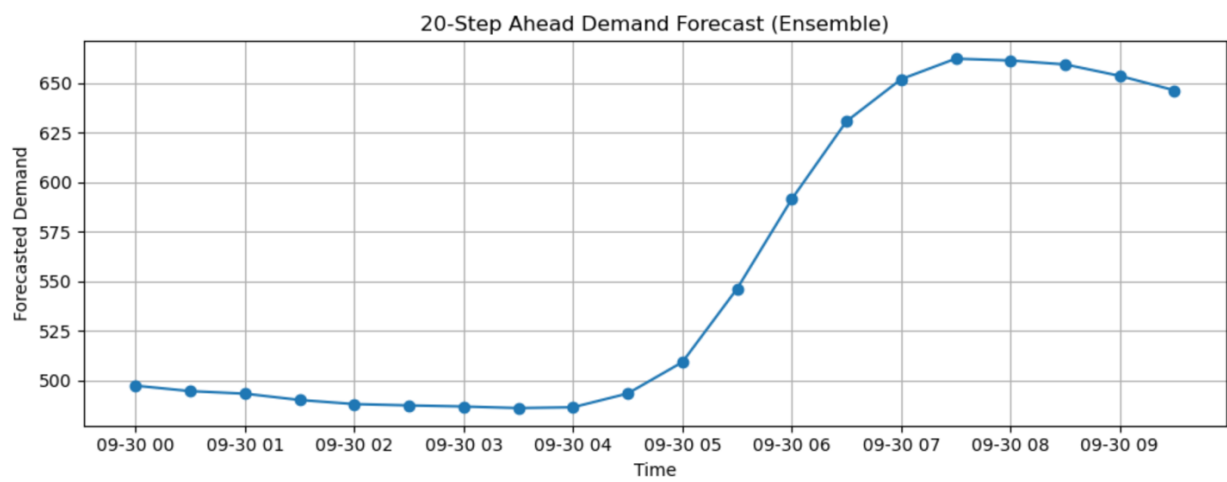


Figure 3

Twenty-step ahead electricity demand forecast generated using the ensemble AI model.

5. Benefits to Academia and Industry

5.1 Benefits to Sustainable Energy Systems

AI-based ensemble forecasting supports sustainability by enabling more efficient integration of renewable energy. Improved forecasting accuracy reduces reliance on fossil-fuel-based reserves, lowers operational emissions, and supports higher renewable penetration.

5.2 Benefits to Industry

For industry stakeholders, ensemble forecasting provides improved reliability, reduced operational costs, and enhanced flexibility management. The use of interpretable and computationally efficient models improves deployability in real-world energy management systems.

5.3 Benefits to Academia

From an academic perspective, this work demonstrates how ensemble learning theory can be translated into practice. It highlights opportunities for future research in hybrid ensemble–deep learning models, uncertainty quantification, and explainable AI (Dong et al., 2025; Cai et al., 2026).

6. Recommendations

Based on the reviewed literature and applied case study, the following recommendations are proposed:

1. Adopt ensemble AI forecasting approaches in renewable-rich energy systems.
2. Integrate robust feature engineering and data preprocessing to enhance reliability.
3. Balance model complexity and interpretability to support practical deployment.
4. Incorporate uncertainty estimation for risk-aware operational decision-making.
5. Strengthen collaboration between academia and industry to accelerate adoption of AI forecasting solutions.

7. Conclusion

This report investigated the application of artificial intelligence in the sustainability sector, focusing on short-term electricity load forecasting. A review of the state-of-the-art AI applications showed that increasing renewable energy penetration introduces significant variability and nonlinearity, making traditional forecasting methods insufficient. Recent studies highlight ensemble and intelligent machine learning approaches as effective solutions for addressing these challenges.

An ensemble-based forecasting framework was analysed and implemented as a practical case study. The results demonstrated that tree-based machine learning models outperform linear regression methods, and that combining multiple learners through ensemble techniques further improves forecasting accuracy and robustness. The ensemble model achieved the lowest prediction error, confirming its suitability for short-term load forecasting in modern power systems.

Accurate AI-based load forecasting enables improved demand–supply balancing, reduced reliance on fossil-fuel-based reserves, and better integration of renewable energy sources. From an academic perspective, the study supports ongoing research into ensemble and hybrid forecasting methods. From an industry perspective, the implemented model demonstrates a deployable and effective approach for operational energy management. Overall, ensemble AI forecasting represents a practical and valuable tool for supporting sustainable power system operation.

References

Cai, J., Cai, Y., Yan, Y., Chen, L., & Zhang, X. (2026). Synergistic cyclic optimization strategy for the data screening and forecasting of solar power, Wind power, and electricity load. *Renewable Energy*, 256, 124524. <https://doi.org/10.1016/J.RENENE.2025.124524>

Dong, Q., Huang, R., Cui, C., Towey, D., Zhou, L., Tian, J., & Wang, J. (2025). Short-Term Electricity-Load Forecasting by deep learning: A comprehensive survey. *Engineering Applications of Artificial Intelligence*, 154, 110980. <https://doi.org/10.1016/J.ENGAPPAI.2025.110980>

Liu, T., Li, F., Zhang, D., Shan, L., Zhu, H., Du, P., Jiang, M., Goh, H. H., Kurniawan, T. A., Huang, C., & Kong, F. (2026). Intelligent load forecasting technologies for diverse scenarios in the new power systems. *Renewable and Sustainable Energy Reviews*, 226, 116366. <https://doi.org/10.1016/J.RSER.2025.116366>

Pramanik, A. S., Sepasi, S., Nguyen, T. L., & Roose, L. (2024). An ensemble-based approach for short-term load forecasting for buildings with high proportion of renewable energy sources. *Energy and Buildings*, 308, 113996. <https://doi.org/10.1016/J.ENBUILD.2024.113996>

Appendix A

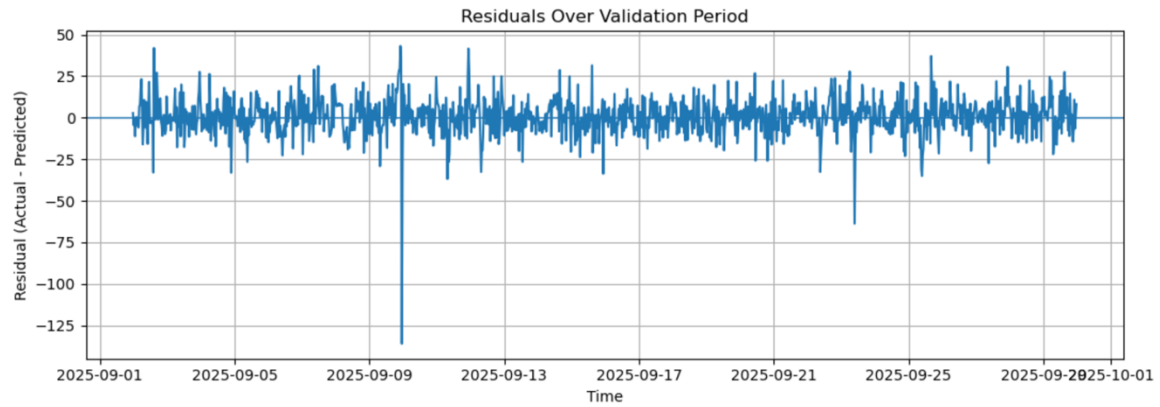


Figure A1

Residuals (actual minus predicted demand) across the validation period for the ensemble forecasting model.

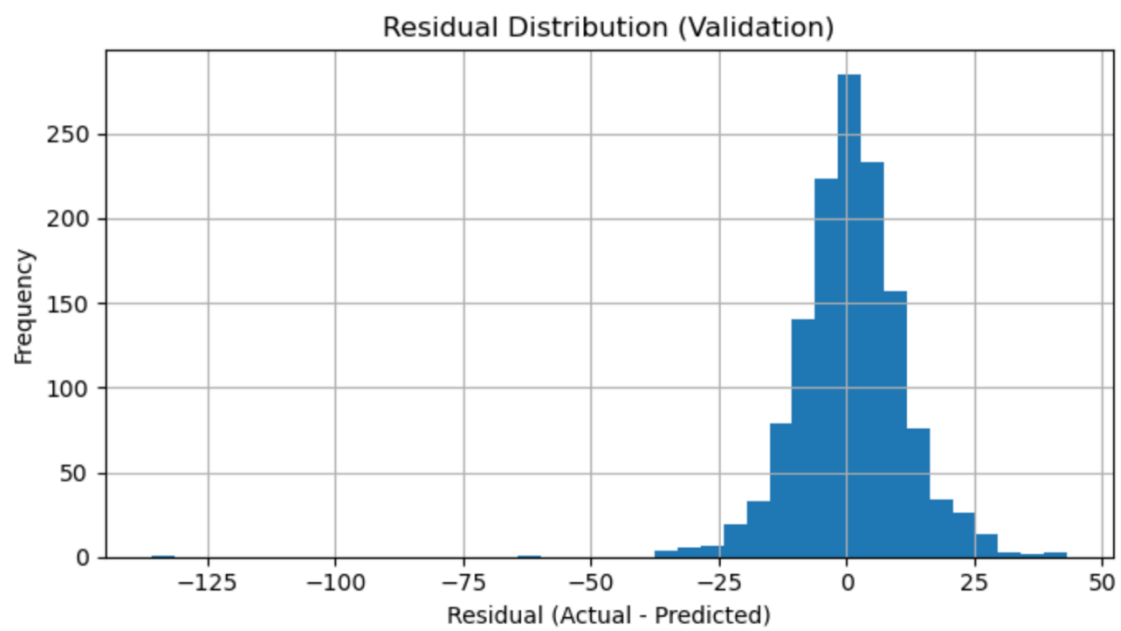


Figure A2

Distribution of ensemble forecast residuals over the validation dataset.

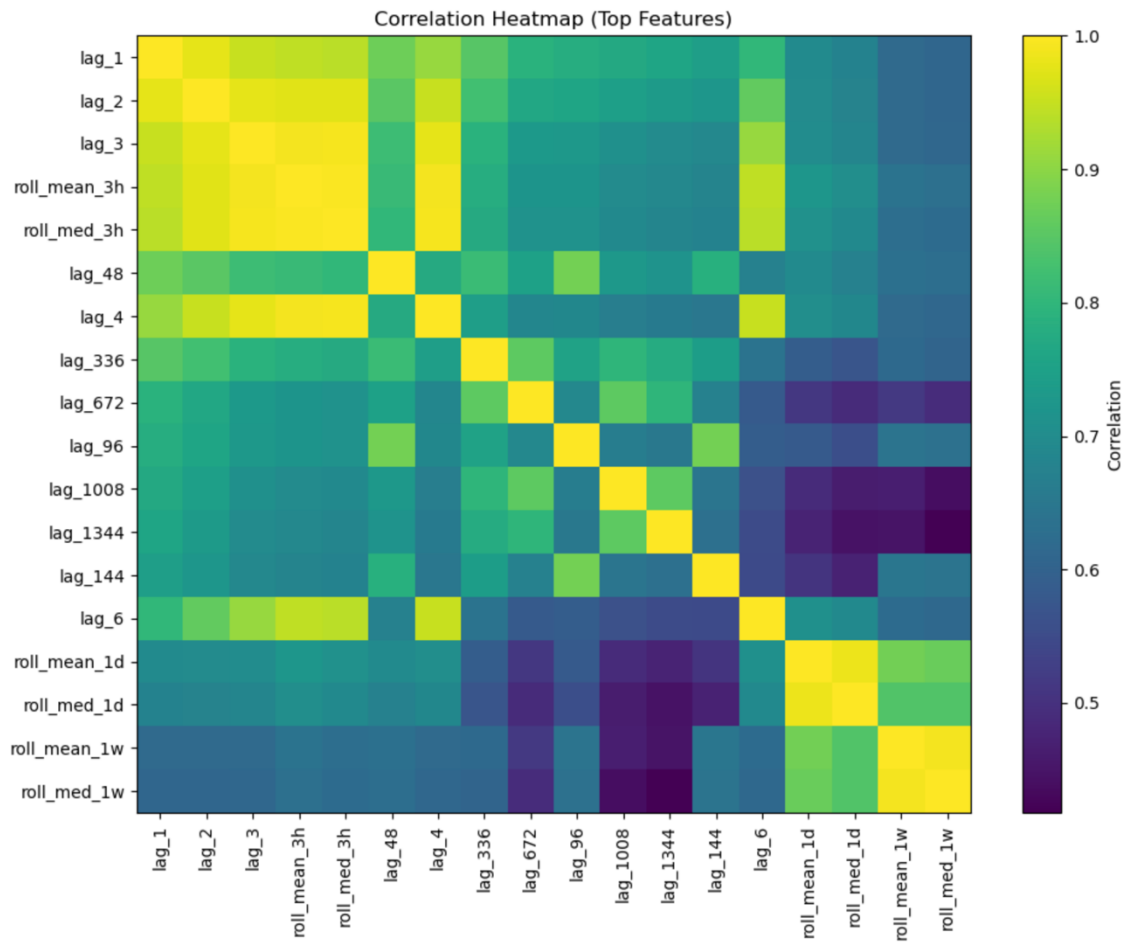


Figure A3

Correlation heatmap of the most influential engineered features used in the forecasting model.

Appendix B

Model Implementation Code

The full implementation of the ensemble forecasting model including data preprocessing, feature engineering, model training, ensemble weight optimisation and result visualisation is provided in the Google Colab notebook available at the following link:

<https://tinyurl.com/55x9a6a7>

The notebook contains all code required to reproduce the results reported in this study.